JEE

Prayas JEE 2025

Maths

Trigonometric Functions

DPP: 4

Q1 The expression $3\left[\sin^4\left(\frac{3\pi}{2}-\alpha\right)+\sin^4(3\pi+\alpha)\right]$ $-2\left[\sin^6\left(\frac{\pi}{2}+\alpha\right)+\sin^6(3\pi+\alpha)\right]$

(A) 0

(B) 1

(C) 3

(D) $\sin 4\alpha+\sin 6\alpha$

Q2 If $\mathrm{A}=\tan 6^{\circ} \tan 42^{\circ}$ and $\mathrm{B}=\cot 66^{\circ} \cot 78^{\circ}$, then

(A) $\mathrm{A}=2\ \mathrm{B}$

(B) $\mathrm{A}=1/3\ \mathrm{B}$

(C) $\mathrm{A}=\mathrm{B}$

(D) $3\ \mathrm{A}=2\ \mathrm{B}$

Q3 $\frac{1}{\cos 290^{\circ}}+\frac{1}{\sqrt{3}\sin 250^{\circ}}=$

(A) $\frac{2\sqrt{3}}{3}$

(B) $\frac{4\sqrt{3}}{3}$

(C) $\sqrt{3}$

(D) None of these

Q4 Proof $\frac{\sin 7A-\sin A}{\sin 8A-\sin 2A}=\cos 4A\sec 5A.$

Q5 Proof $\frac{\cos 2B+\cos 2A}{\cos 2B-\cos 2A}=\cot(A+B)\cot(A-B)$

Q6 Prove that $\frac{\sin 2A+\sin 2B}{\sin 2A-\sin 2B}=\frac{\tan(A+B)}{\tan(A-B)}.$

Q7 Proof : $\frac{\cos 2B-\cos 2A}{\sin 2B+\sin 2A}=\tan(A-B).$

Q8 Proof :

$\cos(\mathrm{A}+\mathrm{B})+\sin(A-B)=2\sin(45^{\circ}+A)\cos(45^{\circ}+B)$

Q9 Proof : $\frac{\cos 3A-\cos A}{\sin 3A-\sin A}+\frac{\cos 2A-\cos 4A}{\sin 4A-\sin 2A}=\frac{\sin A}{\cos 2A\cos 3A}.$

Q10 Proof : $\frac{\sin(4A-2B)+\sin(4B-2A)}{\cos(4A-2B)+\cos(4B-2A)}=\tan(A+B)$

Answer Key

Q1	(B)	Q6	proof
Q2	(C)	Q7	proof
Q3	(B)	Q8	proof
Q4	Proof	Q9	proof
Q5	Proof	Q10	Proof